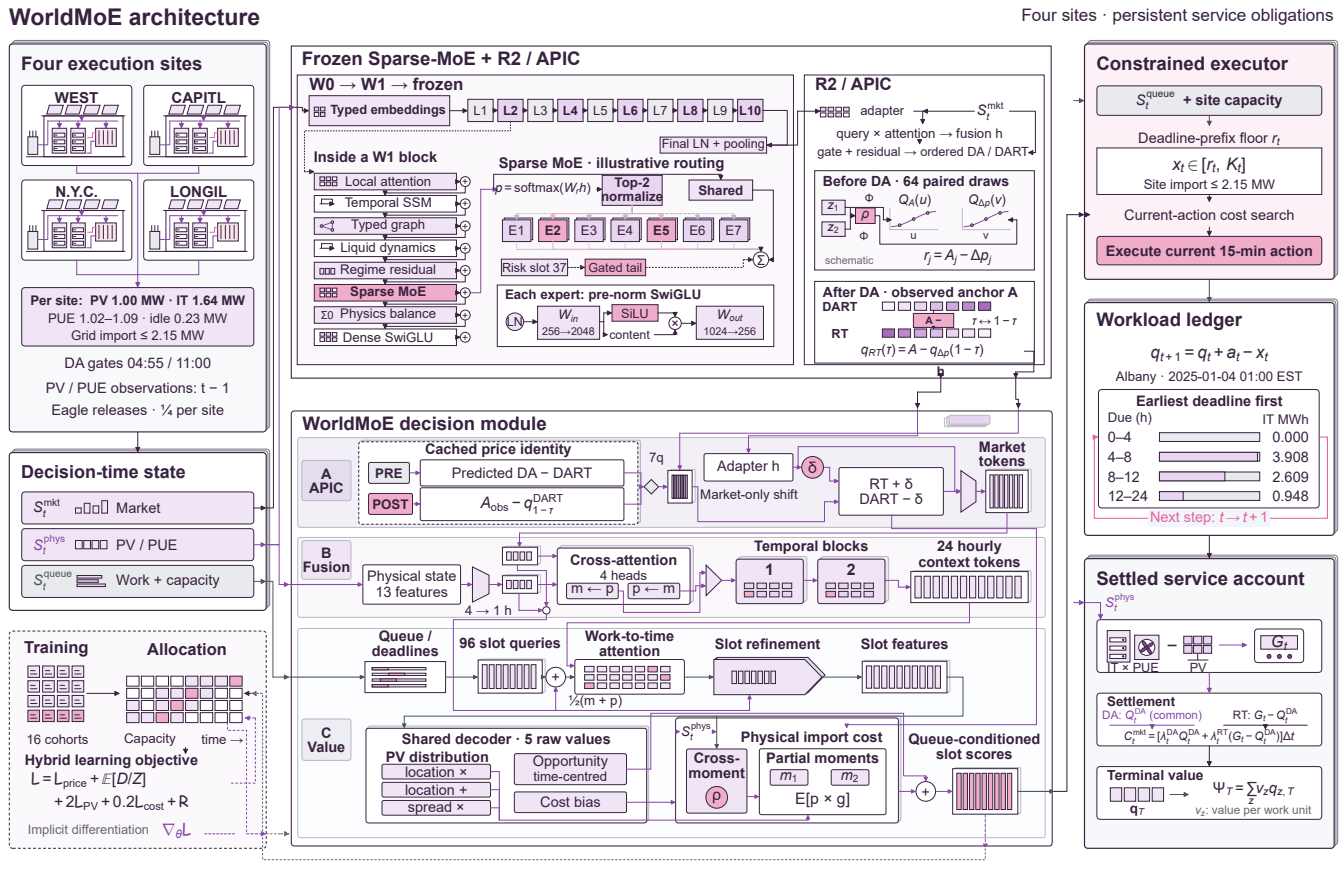

WorldMoE architecture
Four sites · persistent service obligations
Four execution sites
WEST
CAPITL
N.Y.C.
LONGIL
Per site: PV 1.00 MW · IT 1.64 MW
PUE 1.02–1.09 · idle 0.23 MW
Grid import ≤ 2.15 MW
DA gates 04:55 / 11:00
PV / PUE observations: t − 1
Eagle releases · ¼ per site
Decision-time state
Market
PV / PUE
Work + capacity
Training
Allocation
16 cohorts
Capacity
time
Hybrid learning objective
Implicit differentiation
Frozen Sparse-MoE + R2 / APIC
W0 → W1 → frozen
Typed embeddings
L1 L2 L3 L4 L5 L6 L7 L8 L9 L10
Final LN + pooling
Inside a W1 block
Local attention
Temporal SSM
Typed graph
Liquid dynamics
Regime residual
Sparse MoE
Physics balance
Dense SwiGLU
Sparse MoE · illustrative routing
Top-2 normalize
Shared
E1 E2 E3 E4 E5 E6 E7
Risk slot 37
Gated tail
Each expert: pre-norm SwiGLU
SiLU
content
R2 / APIC
adapter
query × attention → fusion h
gate + residual → ordered DA / DART
Before DA · 64 paired draws
schematic
After DA · observed anchor A
DART
RT
WorldMoE decision module
A APIC
Cached price identity
PRE
Predicted DA − DART
POST
7q
Adapter h
Market-only shift
RT + δ
DART − δ
Market tokens
B Fusion
Physical state
13 features
4 → 1 h
Cross-attention
4 heads
Temporal blocks
24 hourly context tokens
Queue / deadlines
96 slot queries
Work-to-time attention
Slot refinement
Slot features
C Value
Shared decoder · 5 raw values
PV distribution
location ×
location +
spread ×
Opportunity time-centred
Cost bias
Physical import cost
Cross-moment
Partial moments
Queue-conditioned slot scores
Constrained executor
+ site capacity
Deadline-prefix floor
Site import ≤ 2.15 MW
Current-action cost search
Execute current 15-min action
Workload ledger
Albany · 2025-01-04 01:00 EST
Earliest deadline first
Due (h)
IT MWh
0–4
0.000
4–8
3.908
8–12
2.609
12–24
0.948
Next step:
Settled service account
Settlement
DA: (common)
RT:
Terminal value